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Fe and N Co-Doped Porous Carbon Nanospheres with High Density of Active Sites for Efficient CO ElectroreductionYubin Chen, Liangliang Zou, Hong Liu, Chi Chen, Qi Wang,
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**Fe and N Co-Doped Porous Carbon Nanospheres with High Density of
Active Sites for Efficient CO₂ Electroreduction**

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Abstract: To enhance the electrocatalytic performance of transition metal and nitrogen co-doped carbon (M-N-C) for CO₂ electro-reduction reaction, we present a Fe and N co-doped porous carbon nanosphere (Fe-N-PC) with high density of single-atom iron active sites and a few Fe clusters, as indicated by high-resolution TEM and EXAFS, using SiO₂ as protecting shell to restrict isolated Fe atoms and to trap the volatile N-containing species. The Fe doping content can reach as high as 3.9 wt. %. The Fe-N-PC catalyst exhibits an excellent performance for converting CO₂ to CO with high faradic efficiency (~90 %) and remarkable partial CO current density (11.44 mA cm⁻²) at -0.49 V. Probing analysis and selective chemical modification reveal that both Fe atom and ortho-C of N atoms play a synergetic role in enhancing CO₂RR activity. DFT calculations reveal that the presence of amorphous Fe clusters can facilitate the desorption of CO from the surface of the catalyst, thus further improving the CO₂RR activity. This work presents a general strategy to synthesize advanced M-N-C material with high density of single atom active sites, which could be used as an efficient electrocatalyst for CO₂RR, ORR et al.

1. Introduction

CO₂ electro-reduction reaction (CO₂RR) has received extensive attentions due to ambient reaction conditions and easily controllable high value-added products. Since the CO₂RR suffers from a high overpotential, it is often accompanied by the hydrogen evolution reaction (HER), leading to a poor selectivity of the target products.¹⁻² In the past few years, a series of catalysts, such as metals,³⁻⁴ metallic oxides,⁵ metal alloys,⁶ etc., have been reported in order to improve the efficiency of CO₂RR. For instance, metallic Au,⁷⁻⁹ Ag¹⁰ or Pd¹¹ catalysts have been found to be able to catalyze electroreduction of CO₂ to CO and HCOOH. However, the insufficient activity and low selectivity, as well as the prohibitive price of these catalysts severely impede the future applications.

Transition metal and nitrogen co-doped carbon (M-N-C) materials have been recently considered as promising substitutes for noble metal catalysts due to their low cost, high surface area, tunable structure, and unique electronic feature. A variety of M-N-C materials, including carbon nanotubes,¹² carbon nanospheres,¹³ and graphene¹⁴, have emerged as electrocatalysts for CO₂RR in recent years. For example, Strasser et al.¹⁵ developed a variety of M-N-C catalysts (M=Mn, Fe, Co, Cu, Ni) to study the effects of transition metal on CO₂RR performance. They have found that the Ni-N-C catalyst presented the highest current for CO formation but at very high potentials (<-0.75 V vs. RHE) and that the Fe-N-C was identified to obviously decrease the overpotential of CO₂RR. As reported, the introduction of metallic Fe nanoparticles within Fe-N-C materials could further reduce the overpotential.¹⁶ Our previous work¹⁷ has also demonstrated that the Fe_xN nanoparticles encapsulated within Fe and N co-doped carbon layers can further promote the activity for the conversion of CO₂ to CO at a low overpotential, but there is a great space for the current improvement. Eventually, low catalytic current for CO

formation from CO₂RR on the Fe-N-C could be ascribed to an insufficient density of the active sites,^{12, 18} resulting from the low Fe and N doped content and the aggregation of Fe atoms. The aggregation of transition metal within M-N-C precursors, usually occurring during the heat-treatment at high temperature, would be favorable for the hydrogen evolution reaction (HER) rather than the CO₂RR.^{16, 18-19} Therefore, it is crucial to develop a strategy that can restrain the aggregation of Fe and increase the density of Fe-N active sites. In the meantime, the Fe-N_x moieties are commonly accepted as the catalytically active sites for the CO₂RR^{16, 18, 20-21}, however, the exact effects of Fe, N and adjacent C atoms are still needed to be clarified. To this end, the understanding of the active sites is of great significance to the development of novel M-N-C catalysts.

In this work, we develop a strategy to synthesize mesoporous Fe and N co-doped porous carbon nanospheres (Fe-N-PC) with high density of atomic Fe active sites and a few Fe clusters, employing SiO₂ as template and protecting shell. Simultaneously, Fe precursors were trapped into polymer network through the electrostatic interaction. SiO₂ shell²²⁻²³ acts as a microreactor to prevent aggregation of isolated Fe atoms and volatilization of N-containing species during the formation of Fe-N-C materials at high temperature. The optimal Fe-N-PC catalyst exhibits excellent activity for electroreduction of CO₂ to CO with high faradic efficiency (FE) of 90 % and partial CO current density of 11.44 mA cm⁻² at -0.49 V versus reversible hydrogen electrode (RHE). The poisoning experiment and selective chemical modification verify that both Fe atom and ortho-C of pyridinic N are conducive to improve the CO₂RR activity. Theoretical calculations demonstrate that amorphous Fe clusters can dramatically lower CO bonding strength, thus facilitating the CO₂RR activity of Fe-N_x active sites.

2. Experimental Section

2.1. Synthesis of Fe-N-PC

1.5 g of SiO₂ nanosphere (25 nm) was dispersed in 35 mL of deionized water, and then 900 μ L of pyrrole was injected into the SiO₂ suspensions under vigorous stirring. After that, 15 mL of aqueous solution containing 0.65 g of ammonium thiosulphate ((NH₄)₂S₂O₄) and 0.78 g of potassium ferricyanide was poured into the above mixture under stirring and reacted in ice bath for 24 h. After centrifugation, washing with water and drying, the obtained SiO₂@PPy-Fe(CN)₆ was dispersed in 90 mL of ethanol under sonication and vigorous stirring, followed by injection of 10 mL of ammonium hydroxide (NH₃·H₂O, 28 wt. %). Then, 4 mL of tetraethylorthosilicate (TEOS) was dropwise added to the mixture and reacted at room temperature for 24 h. The product (SiO₂@PPy-Fe(CN)₆@SiO₂) was collected by rotary evaporation and ground into fine powders. The precursors were then pyrolyzed at 900 °C for 2 h with a heating rate of 5 °C min⁻¹ under N₂ protection. After cooling to room temperature, the sample was immersed in 4 mol·L⁻¹ NaOH solution under stirring at 80 °C for 24 h to etch the SiO₂, followed by centrifugation, washing with 1 mol·L⁻¹ NaOH and water several times and vacuum drying at 60 °C. Finally, the sample was annealed to 900 °C under N₂ protection with a rate of 5 °C min⁻¹ and held for 10 min in NH₃ with a flow rate of 200 mL min⁻¹. Then the sample was cooled to room temperature protected by N₂ naturally to get the Fe-N-PC.

For a comparison, the samples without the addition of NH₃·H₂O and TEOS to form SiO₂ shell (denoted as op-Fe-N-PC) and without the addition of Fe (denoted as N-PC) were also synthesized.

To achieve selective chemical modification, the best Fe-N-PC (100 mg) was dispersed in 5 mL of ethanol and 5 mL of 40 % aldehyde aqueous solution. 100 mg of [Bis(trifluoroacetoxy)iodo]-benzene was added and the mixture was stirred under room

temperature for 48 h. The precipitates were separated by centrifugation and washing with ethanol and water. The product is denoted as modified Fe-N-PC.

To adjust the Fe content within the catalysts, different amounts of $\text{K}_3\text{Fe}(\text{CN})_6$ were added in synthetic route. 0.26, 0.52, 0.68 and 0.78 g $\text{K}_3\text{Fe}(\text{CN})_6$ were used to synthesize Fe-N-PC-1, Fe-N-PC-2, Fe-N-PC-3 and Fe-N-PC-4.

2.2. Physical characterizations

The morphologies were characterized by transmission electron microscopy (TEM, FEI Tencnai 30F, 200 kV) and high angle annular dark field scanning transmission electron microscopy (HADDF-STEM, FEI Titan Themis G2 60-300). The crystalline structure was characterized by X-ray diffraction (XRD, Bruker AXS D8 ADVANCE). The chemical states and compositions were measured by X-ray photoelectron spectroscopy (XPS, Thermo Fisher Escalab 250 Xi). The surface areas and porosities were determined by measuring the N_2 adsorption-desorption isotherms on Micromeritics ASAP-2020 instrument. The X-ray absorption fine structure (XAFS) analysis was conducted on BL14W1 beamline of Shanghai synchrotron radiation facility. Liquid product was characterized by ^1H NMR on Bruker AVANCE AV III 400. Function group of sample was characterized by fourier transform infrared spectrometer (FTIR, Thermo Fisher Nicolet IS5).

2.3. Electrochemical measurements

All the electrochemical measurements were performed on a CHI 760E workstation with a three-electrode system. To prepare the working electrode, 10 mg of the Fe-N-PC catalyst was dispersed in the mixture of 0.05 mL of 5 wt. % Nafion and 0.95 mL of ethanol by sonication for 30 min to form homogeneous ink. 20 μL of the ink was pipetted onto the pre-polished glassy carbon (GC) electrode with a diameter of 5 mm, leading to a loading of 1 mg cm^{-2} . A glassy

carbon plate and Ag/AgCl electrode were used as the counter electrode and the reference electrode. All the linear sweep voltammetry (LSV) measurements were recorded in CO₂-saturated and N₂-saturated 0.5 mol·L⁻¹ KHCO₃ electrolyte. All the potentials were converted to the range of reversible hydrogen electrode (RHE) according to $E(\text{RHE}) = E(\text{Ag/AgCl}) + 0.205 \text{ V} + 0.0591 \times \text{pH}$. The pH of CO₂-saturated 0.5 mol·L⁻¹ KHCO₃ is 7.2.

Controlled potential electrolysis (CPE) method was performed on a H-shape electrolysis cell with a Nafion 115 membrane as the separator using three-electrode system. The porous working electrode was prepared by the gas-diffusion electrode (GDE) method. In brief, the XC-72 carbon powder, polytetrafluoroethylene (PTFE) colloid and isopropanol/water (IPA/H₂O) were mixed with the weight ratio of 4:1:120. The suspension was sprayed on the carbon paper to form the micro-porous layer (MPL) with the carbon loading of 1 mg cm⁻². Thereafter, the as-prepared catalysts and Nafion resin with the mass ratio of 3:1 were ultrasonically dispersed in IPA/H₂O solution for 3 h and then coated on the MPL with the catalyst loading of 1.0 mg cm⁻². The geometric reaction area of electrode was 1×1 cm². The electrolyte was CO₂-saturated 0.5 mol·L⁻¹ KHCO₃ solution. During the experiment, CO₂ was continuously bubbled into the cathode at a constant rate of 20 mL min⁻¹. The gaseous products of cathodic compartment were directly vented into a gas chromatograph (GC) equipped with a thermal conductive detector (TCD) and a flame ionization detector (FID). Each electrode was electrolyzed under applied potentials for 20 min. And the GC run was initiated every 10 min. The faradaic efficiencies (FE) of the gas products were calculated by the following equation:

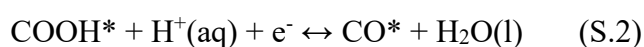
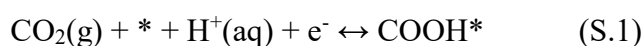
$$FE \% = \frac{C \times 10^{-6} \times V_{\text{CO}_2} \times 10^{-3} \times 60 \times 96485 \times n}{22.4 \times Q}$$

Where V_{CO_2} is the flow rate of CO₂ (20 mL min⁻¹), C (ppm) is the concentration of the gaseous products (CO or H₂) detected by GC, n is the transfer electrons for producing CO or

H₂, and Q (A·s) is the total electric charge.

2.4. Density functional theory (DFT) calculation

DFT calculations were performed using the Vienna Ab initio Simulation Package (VASP) code²⁴ with projector augmented wave (PAW) method.²⁵⁻²⁸ The generalized gradient approximation (GGA) with Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional was applied to set the plane wave basis.²⁹ The Monkhorst–Pack mesh k-point grids was 3×3×1 for all supercells Brillouin zone. The plane wave cutoff was 500 eV and the atomic structures were allowed to relax until residual force were below 0.05 eV/Å. In order to separate the slab from its periodic image in the normal direction, the vacuum thicknesses were set more than 12 Å. As for electrochemical CO₂RR, we considered the following elementary steps:



where (g), * and (aq) represent the gaseous phase, adsorption state and aqueous phase, respectively. Reaction free energy change of each elementary step was estimated by using the computational hydrogen electrode model.³⁰ The free energies of each species were calculated according to the formula:

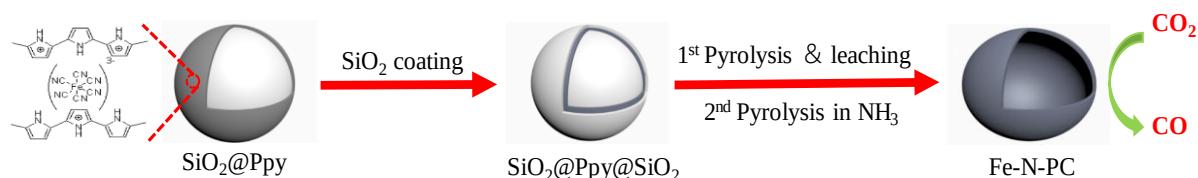
$$G = E_{\text{DFT}} + E_{\text{ZPE}} + \int C_p dT - TS + E_{\text{sol}} \quad (\text{S.4})$$

where E_{DFT} is the DFT calculated total energy, E_{ZPE} is the zero point energy, C_p is the constant pressure heat capacity and S is the entropy, the E_{ZPE} , C_p and S are based on the literature that has been reported.³¹ The solvation effect was considered using the values reported in the literature that CO* was stabilized by 0.1 eV and COOH* by 0.25 eV.³² The errors of PBE

calculations for gas molecules were corrected with +0.13 eV, -0.51 eV, -0.08 eV and -0.06 eV for CO₂, CO, H₂ and H₂O, respectively.³²

3. Results and discussion

The synthesis of Fe-N-PC is schematic illustrated in Scheme 1. Firstly, the polymerization of pyrrole monomers is triggered with (NH₄)₂S₂O₄ as oxidant to form PPy layers coating on SiO₂ nanosphere. The Fe(CN)₆³⁻ dissolved in the solution can be assembled into PPy network through the electrostatic interaction because of the positively charged groups of PPy chain. The coating of outer SiO₂ shell was prepared by hydrolyzing TEOS with NH₃·H₂O³³, which is able to prevent the migration of Fe atoms and volatilization of N-containing species. After pyrolysis at 900 °C under N₂, etching the SiO₂ with NaOH and second heat-treatment under NH₃, the final Fe-N-PC catalysts can be obtained with the porous nanosphere structure. For the sake of comparison, the op-Fe-N-PC without the protection of SiO₂ shell was also synthesized.



Scheme 1. Schematic diagram for the preparation of Fe-N-PC catalyst.

As reported in the literature^{12, 18}, metal within the M-N-C catalyst is at least the part of the active center and its content is crucial for the improvement of CO₂RR. In this work, we keep the N content constant (Table S1) and adjust the Fe content within the catalyst. As can be seen in Figure S1, with the increase in Fe content within the Fe-N-C catalysts, the activity and selectivity of CO₂RR to CO are significantly improved, assessing the importance of Fe content. The Fe content of the best-performing Fe-N-PC is as high as 3.9 wt. %, which is higher than the most Fe-N-C catalysts reported to date (Table S2). In this case, we can attribute the observed high-performance of Fe-N-PC catalyst for CO₂RR to high density of active sites.

The morphology of the as-synthesized catalysts was firstly characterized by TEM. As

shown in Figure 1a and Figure S2, the Fe-N-PC shows the structure of defective porous nanospheres with a diameter of ~50 nm. Furthermore, no obvious Fe-containing nanoparticles could be observed from the TEM images of Fe-N-PC. The aberration-corrected HAADF-STEM images (Figure 1b and Figure S3) of Fe-N-PC reveal that the Fe atoms are mostly dispersed as isolated single atoms throughout the carbon substrate, which are identified by bright dots marked with red circles. It is worth mentioning that the bright areas (1~2 nm) could be attributed to the amorphous Fe clusters. The elemental mapping analysis (Figure 1c) reveals that the Fe and N atoms are homogeneously distributed over the whole carbon matrix.

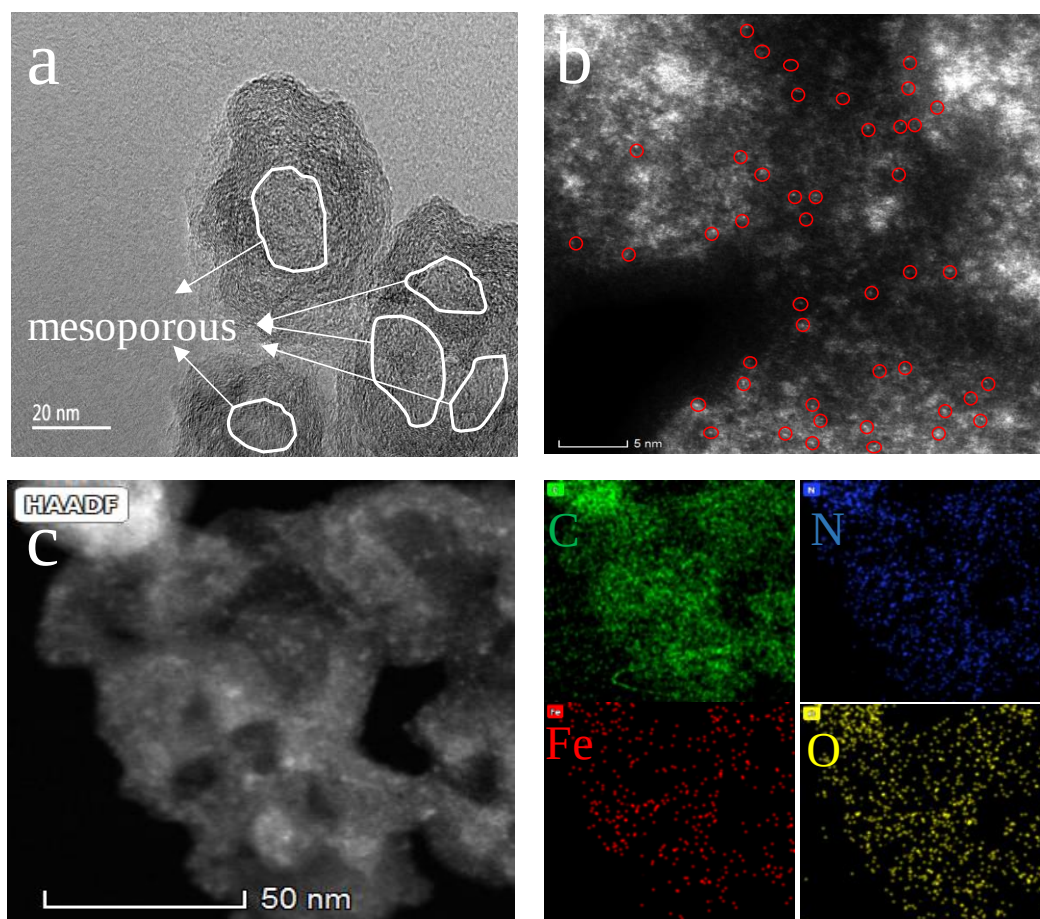


Figure 1. (a) TEM image of Fe-N-PC, mesoporous are marked with white curves; (b) aberration-corrected HAADF-STEM image of Fe-N-PC; (c) elemental mapping analysis of Fe-N-PC and the distributions of C, N, Fe and O elements.

X-ray adsorption near edge structure (XANES) and extended X-ray adsorption fine structure (EXAFS) spectroscopy were carried out to unravel the chemical status and coordination environment of Fe in Fe-N-PC. The XANES curves of Fe-N-PC and op-Fe-N-PC show near-edge absorption energy between those of iron(II) phthalocyanine ($\text{Fe}^{\text{II}}\text{Pc}$) and Fe foil, implying that Fe atoms carry positive charges (Figure 2a). As shown in Figure 2b, the experimental Fourier transform (FT) at Fe K-edge EXAFS exhibits an obvious peak at ~ 1.5 Å for both Fe-N-PC and op-Fe-N-PC, which can be attributed to Fe-N(O) scattering paths, matched well with the peak of $\text{Fe}^{\text{II}}\text{Pc}$. Interestingly, the EXAFS spectrum of Fe-N-PC displays a weak peak at around 2.1 Å, which can be assigned as Fe-Fe bond. Combined with the HAADF-STEM observation, a few Fe clusters might exist in Fe-N-PC, which are probably formed during NH_3 heat-treatment. By contrast, the FT-EXAFS curve of the op-Fe-N-PC sample presents a high peak at 2.1 Å, indicative of the presence of much more Fe-Fe bond. These results clearly demonstrate the critical role of SiO_2 shell in restricting isolated Fe atoms from aggregation, hence greatly improving the density of atomic Fe active sites.

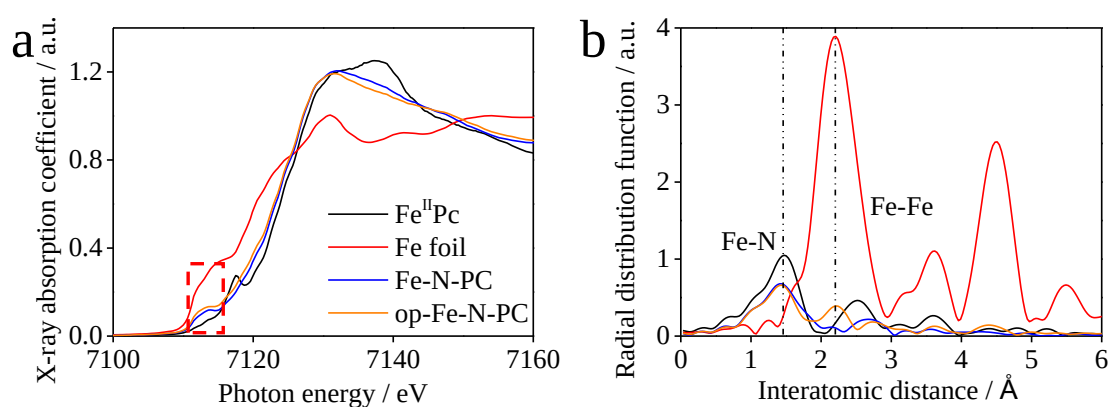


Figure 2. (a) Fe K-edge XANES spectra and (b) Fourier-transformed module of EXAFS signal of Fe-N-PC, op-Fe-N-PC and the reference samples ($\text{Fe}^{\text{II}}\text{Pc}$, Fe foil).

XRD measurements were used to characterize the crystalline structure of the Fe-N-PC

catalysts. Without the protection of SiO₂ shell, the metallic Fe nanoparticles are formed during the first heat-treatment under N₂ (Figure S4). However, with SiO₂ shell, no characteristic peaks for metallic Fe phases could be observed in the XRD pattern of Fe-N-PC (Figure 3a). The two broad peaks at 26° and 44° can be assigned to the reflections of (002) and (101) planes of graphitic carbon. Without the use of SiO₂ shell, the XRD pattern of op-Fe-N-PC display a sharp peak at 44.8°, indicating the formation of metallic Fe nanoparticles, which is consistent with the EXAFS results.

The Brunauer-Emmett-Teller (BET) surface areas and porosities of the Fe-N-PC catalysts were quantified by measuring N₂ adsorption-desorption isotherms (Figure 3b). Both Fe-N-PC and op-Fe-N-PC samples show a type IV sorption isotherm. The adsorption volume at a low relative pressure confirms the existence of micropores, while the distinct hysteresis loop in the range of $P/P_0=0.5-0.8$ is the representation of mesoporous.³⁴ The BET specific surface area and micropore area of Fe-N-PC catalyst are 579.5 and 111.6 m² g⁻¹, respectively, which are higher than that of op-Fe-N-PC (430 and 85.9 m² g⁻¹). The pore size distribution curves (inset in Figure 3b) reveal that the pore diameter of Fe-N-PC and op-Fe-N-PC is ~20 nm, which is consistent with the SiO₂ template (Figure S5). The difference in both BET specific surface area and micropore area for those catalysts might be due to the formation of Fe nanoparticles on op-Fe-N-PC.²²

XPS was conducted to probe the composition and chemical states of Fe-N-PC. As shown in Figure 3c, the survey spectra of Fe-N-PC and op-Fe-N-PC display four peaks corresponding to Fe 2p (~710.3 eV), C 1s (284.8 eV), N 1s (~398.8 eV) and O 1s (~532.6 eV), verifying the co-doping of Fe and N within carbon matrix. The N doping content of Fe-N-PC is 7.82 at. %, while the Fe content reaches as high as 3.9 wt. %, determined by inductively coupled plasma

optical emission spectrometer (ICP-OES), as summarized in Table S1. The deconvoluted high-resolution Fe 2p XPS spectrum of Fe-N-PC displays four peaks centered at 710.1, 712.5, 723.2, and 726.0 eV, which are assigned to $\text{Fe}^{2+} 2p_{3/2}$, $\text{Fe}^{3+} 2p_{3/2}$, $\text{Fe}^{2+} 2p_{1/2}$ and $\text{Fe}^{3+} 2p_{1/2}$, respectively. The peak of metallic Fe (707.0 eV) is not found, further proving that the most of Fe exists in the form of single atoms (Figure S6). In addition, the N content of Fe-N-PC (7.82 at. %) is much higher than that of op-Fe-N-PC (4.75 at. %). The high-resolution N 1s XPS spectra of Fe-N-PC and op-Fe-N-PC can be deconvoluted into four peaks with binding energies of 398.5, 399.7, 401, and 403.5 eV, corresponding to pyridinic N, pyrrolic N, graphitic N, and oxidized N, respectively, as shown in Figure 3d.³⁵ Notably, the peak of pyridinic N may also include the contribution of the Fe–N species on account of the small difference in binding energies between Fe-N and pyridinic N.³⁶ Obviously, the pyridinic N and graphitic N occupy the dominant percentage of N configurations, which would facilitate the catalytic activity.³⁷⁻³⁸ The percentages of different N configurations of op-Fe-N-PC are similar with those of Fe-N-PC. The XPS results further indicate that SiO_2 protecting shell prevents not only the aggregation of Fe atoms, but also the volatilization of N-containing species during the pyrolysis.

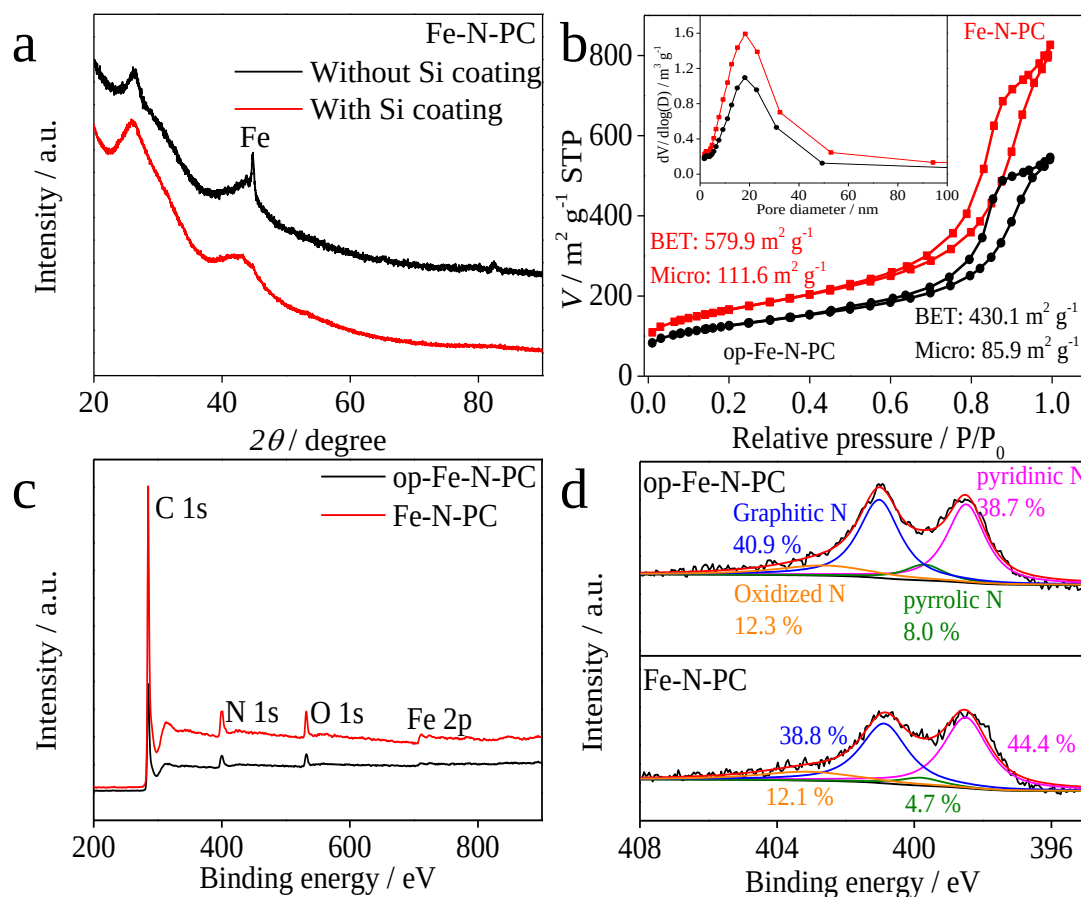


Figure 3. (a) XRD patterns, (b) N₂ adsorption-desorption isotherms (inset shows the pore size distributions), (c) XPS survey spectra and (d) high-resolution N 1s XPS spectra of the Fe-N-PC and op-Fe-N-PC samples.

The CO₂RR catalytic performance of catalysts was firstly evaluated by the linear sweep voltammetry (LSV) with the rotating disk electrode (RDE). The N-PC was also annealed in NH₃ for 10 min to guarantee the similar surface area and nitrogen configurations (Figure S7). The Faradic currents collected in N₂-saturated 0.5 mol·L⁻¹ KHCO₃ (pH=8.8) and CO₂-saturated 0.5 mol·L⁻¹ KHCO₃ (pH=7.2) are resulted from the HER (Figure S8) and the sum of HER and CO₂RR, respectively (Figure 4a). Apparently, the Fe-N-PC catalyst is capable to catalyze the electroreduction of CO₂ according to the much higher current density in CO₂-saturated electrolyte than that in N₂-saturated electrolyte. The catalytic activity of Fe-N-PC for CO₂RR

is remarkably higher than that of Fe-free N-PC, as judged by onset potential and current density, verifying the critical role of Fe atoms for the activity enhancement.³⁹ Furthermore, the CO₂RR catalytic activity of Fe-N-PC is superior to that of op-Fe-N-PC, while the HER activity is lower than that of op-Fe-N-PC, probably because the Fe nanoparticles prefer to catalyze the HER rather than the CO₂RR.^{16, 18-19}

In order to explore the selectivity of the catalysts, electrochemical activities of Fe-N-PC for CO₂ reduction were further examined by controlled-potential electrolysis (CPE) method in CO₂-saturated 0.5 mol·L⁻¹ KHCO₃. Figure 4b shows the partial CO current densities (j_{CO}) of Fe-N-PC, op-Fe-N-PC and N-PC at different potentials. The main products of CO₂RR on all three catalysts are CO and H₂ with trace amount of CH₄ (Figure S9). The sum of CO and H₂ faradic efficiencies (FE_{CO} and FE_{H2}) approaches ~100% (Figure 4c). No liquid products can be detected by ¹H nuclear magnetic resonance after reduction processes (Figure S10). Apparently, the FE_{CO} and FE_{H2} mainly depend on the potentials, because of the competitive nature of CO₂RR and HER. The Fe-N-PC exhibits the highest FE_{CO} and activity at a wide potential range of -0.34~-0.68 V. The current density on Fe-N-PC in this work is nearly four times that of our previous work.¹⁷ The FE_{CO} even reaches over 90 % with a high j_{CO} of 11.44 mA cm⁻² at -0.49 V (RHE) (Figure 4d), superior to that of op-Fe-N-PC with FE_{CO} of 69 % and j_{CO} of 5.49 mA cm⁻² at the same potential. The N-PC shows the worst activity and selectivity for CO formation. The superiorities of SiO₂ protecting shell in facilitating the CO₂RR activity can be ascribed to the high density of atomic Fe-N_x active sites, as demonstrated by HAADF-STEM, EXAFS and XPS results. The activity of the Fe-N-PC is much better than the most Fe-N-C catalysts reported to date (Table S2).

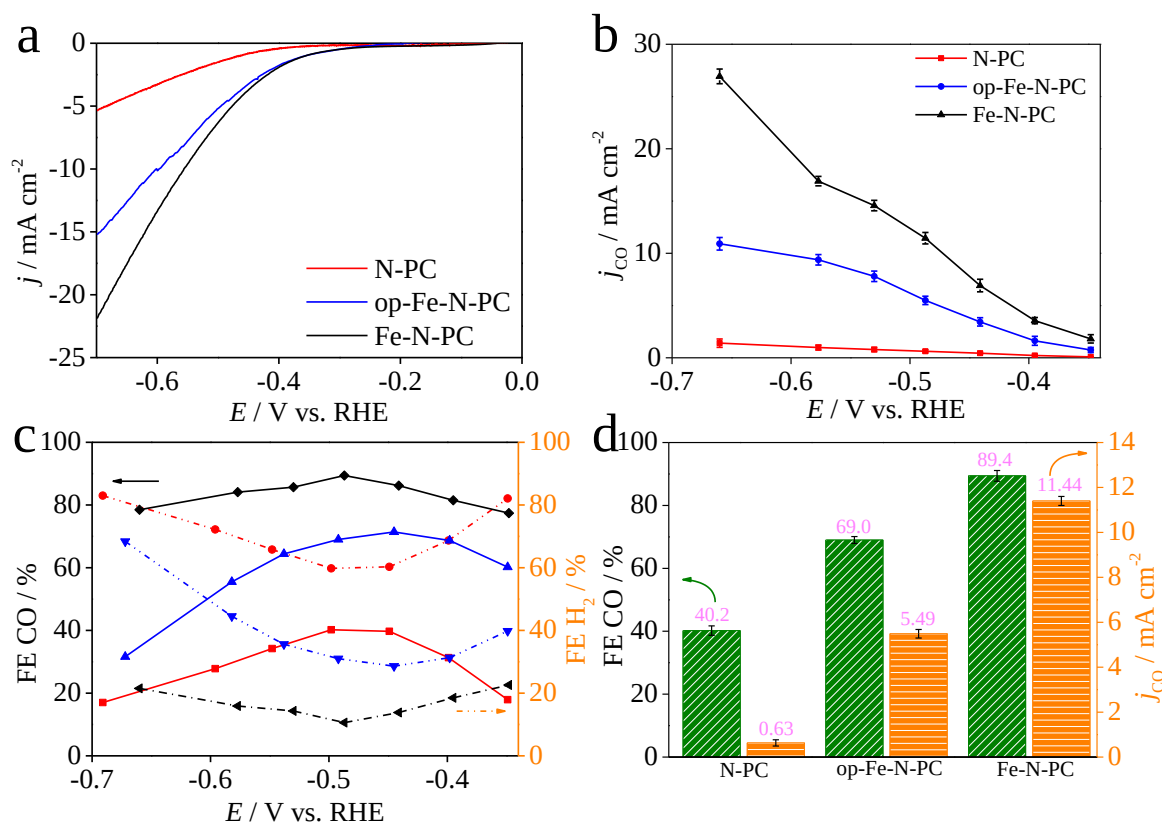


Figure 4. (a) CO₂RR LSV curves of Fe-N-PC, op-Fe-N-PC and N-PC in CO₂-saturated 0.5 mol·L⁻¹ KHCO₃ solution with a scan rate of 10 mV s⁻¹; (b) partial CO current densities and (c) faradic efficiencies of CO/H₂ production of Fe-N-PC, op-Fe-N-PC and N-PC; (d) The comparison of faradic efficiency and current density of CO formation at -0.49 V on various catalysts.

In order to identify the nature of CO₂RR active sites of Fe-N-PC, the poisoning experiments were firstly conducted to probe the effect of Fe-N_x moieties employing thiocyanate ions (SCN⁻), which is widely known for the strong affinity with iron.⁴⁰⁻⁴¹ As shown in Figure 5a, Fe-N-PC exhibits a sharp decrease in current density after injecting 20 mmol KSCN into electrolyte. The FE_{CO} of poisoned Fe-N-PC is only 69.2 % at -0.49 V, accompanied by declined CO₂RR current density and increased FE_{H₂} at overall potentials (Figure 5b). Interestingly, after rinsing the electrode and substituting fresh CO₂-saturated 0.5 mol·L⁻¹ KHCO₃, the CO₂RR

activity of Fe-N-PC recovers to the original level, due to the recovery of the Fe-N₄ sites, clearly manifesting that the Fe atoms at least take part in the CO₂RR active sites.

Exception for the Fe atoms, the pyridinic N and graphitic N have been demonstrated to be crucial for CO₂RR.³⁷⁻³⁸ In addition, the role of ortho-C atom adjacent to N for CO₂RR in Fe-N-PC was further investigated by selective chemical modification of acetyl (denoted as modified Fe-N-PC) through free radical reactions.⁴² XPS and infrared spectroscopy (IR) were carried out to examine the successful modification of modified Fe-N-PC. The IR spectrum of modified Fe-N-PC shows two well-defined peaks at 1630 and 1700 cm⁻¹, as depicted in Figure 5c. The peak at 1700 cm⁻¹ can be assigned to C=O of acetyl group attached on the ortho-C of pyridinic ring, which is similar with the model molecule of 1-(Quinolin-2-yl)ethenone.⁴² The high-resolution N 1s XPS spectra reveal that the peak of pyridinic N of modified Fe-N-PC slightly shifts from 398.5 to 398.8 eV after the modification of ortho-C (Figure 5d), which is also consistent with 1-(quinolin-2-yl)ethanone.⁴² Surprisingly, the CO₂RR activity of modified Fe-N-PC is greatly suppressed as compared with that of Fe-N-PC (Figure 5e), while the HER activity increases significantly (Figure S11). The current density and FE_{CO} of modified Fe-N-PC tested by CPE also demonstrate the severe degradation of CO₂RR activity, as revealed in Figure 5f. This is a direct evidence that in spite of the decisive role of Fe-N₄ moieties for CO₂RR, the ortho-C adjacent to N is also indispensable. The intact active sites probably include the Fe-N₄ center and the surrounding carbon atoms.

Beyond the single atom Fe-N₄-C moieties, the significance of Fe clusters in CO₂RR was also investigated by immersing Fe-N-PC in 1 mol·L⁻¹ H₂SO₄ at 80 °C under stirring for 7 days to remove Fe clusters, denoted as Fe-N-PC-A. The XANES and EXAFS spectra (Figure S12) reveal the existence of Fe-N species, while the weak signal of Fe-Fe in Fe-N-PC disappears,

suggesting the thorough removal of Fe clusters. The CO₂RR catalytic activity and selectivity on Fe-N-PC-A (Figure S13 and S14) decrease to j_{CO} of 6.97 mA cm⁻² and FE_{CO} of 78.4 % at -0.49 V, inferior to that of Fe-N-PC. XPS survey shows that there is almost no change in N structure and content within the Fe-N-PC-A (Figure S15), but a slight decrease in Fe content, probably resulting from the removal of Fe clusters during the acid-leaching (Table S1). Accordingly, the presence of a few Fe clusters may facilitate the CO₂RR catalytic performance.

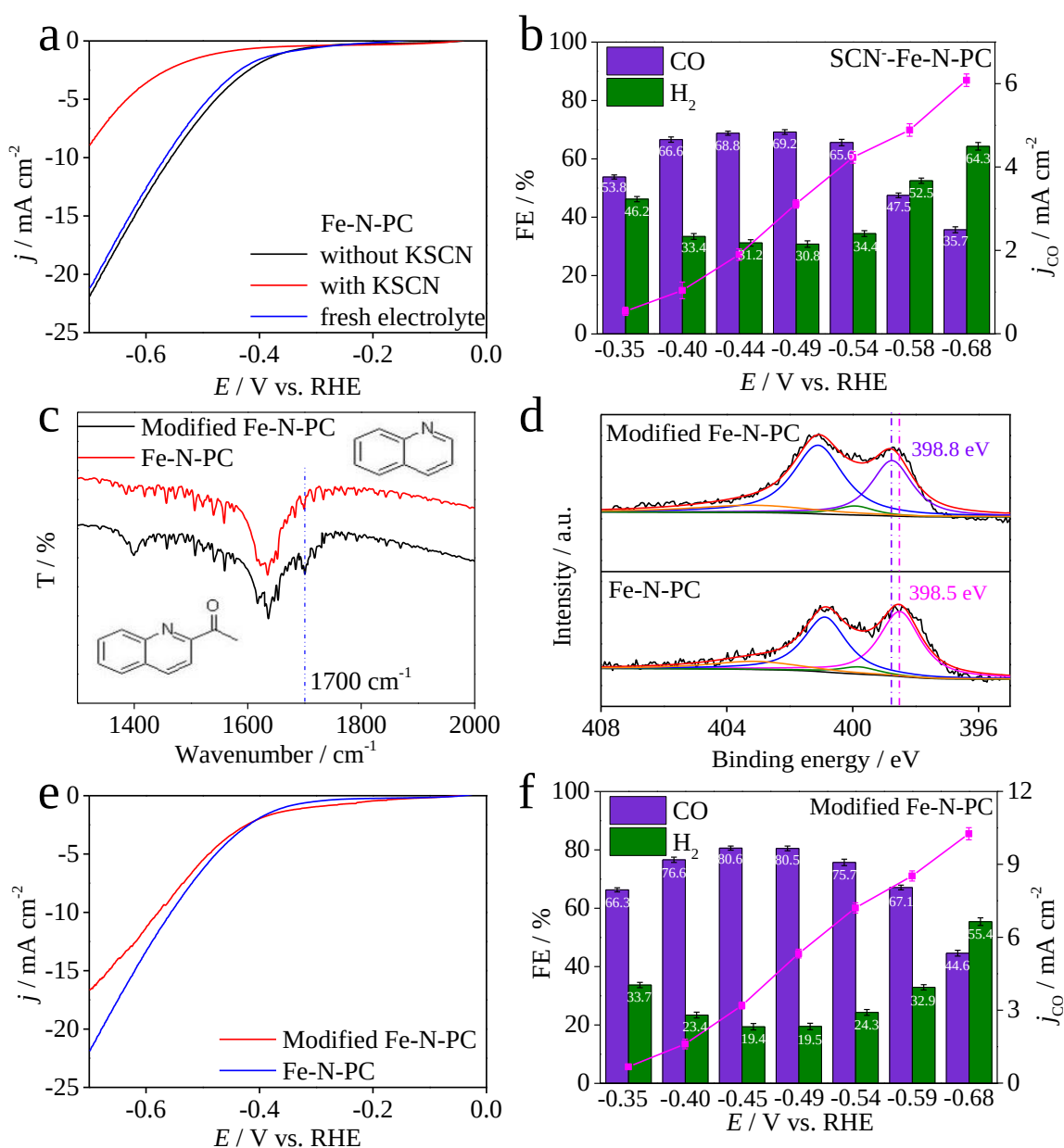


Figure 5. (a) LSV curves of Fe-N-PC in CO₂-saturated 0.5 mol·L⁻¹ KHCO₃ solution with and

without 20 mmol KSCN and after replacing fresh electrolyte; (b) the FE_{CO} , FE_{H_2} and corresponding partial CO current densities of Fe-N-PC after injecting 20 mmol KSCN; (c) IR spectra and (d) high-resolution N 1s XPS spectra of Fe-N-PC and modified Fe-N-PC; (e) LSV curves of Fe-N-PC and modified Fe-N-PC in CO_2 -saturated $0.5 \text{ mol} \cdot \text{L}^{-1} \text{ KHCO}_3$ solution; (f) the FE_{CO} , FE_{H_2} and corresponding partial CO current densities of modified Fe-N-PC.

To understand the synergistic effects of Fe-N₄-C moieties and Fe clusters, DFT calculations were carried out to obtain the free energy diagrams of CO₂RR and HER over Fe-N-PC and Fe-N-PC-A catalysts. The structure with Fe-N₄ embedded in a graphitic plane was built to model the Fe-N-PC-A catalyst, based on the experimental characterization results. The Fe-N-PC catalyst was modeled by adding three extra iron atoms binding with the single Fe atom in Fe-N-PC-A forming a tetrahedron. The structural models for Fe-N-PC and Fe-N-PC-A catalysts are shown in Figure S16. The electroreduction of CO₂ to CO contains three elementary steps and two important intermediates, which are COOH* and CO*. The free energy profiles of the CO₂RR process at 0 V (vs. RHE) are plotted in Figure 6a. It is obvious that the free energy change for the formation of COOH* over Fe-N-PC-A is almost identical with that over Fe-N-PC, and the reaction free energy change for the formation of CO* from COOH* is downhill on both catalyst models. However, the main difference between these two catalysts lies in the CO desorption step. This step is much more difficult on the Fe-N-PC-A catalyst than on the Fe-N-PC catalyst, and more importantly, the desorption of CO is almost free energy neutral on the latter surface. Free energy profiles for CO₂RR at -0.5 V (vs. RHE), which is within the potential range used in our experiment, are shown in Figure 6b. One can see that CO desorption is the rate-determining step at this potential, and the Fe-N-PC catalyst presents better CO₂RR performance. The competition between CO₂RR and HER was also considered here,

and the corresponding adsorption free energies of hydrogen (G_H) on two models were shown in Figure 6c. The HER is much less active on Fe-N-PC than on Fe-N-PC-A. The free energy changes of each elementary step from DFT under 0 and -0.5 V are also listed in Table S3 and S4. Therefore, the three extra iron atoms with Fe-N structure forming a cluster enhance the activity and selectivity of CO₂RR.

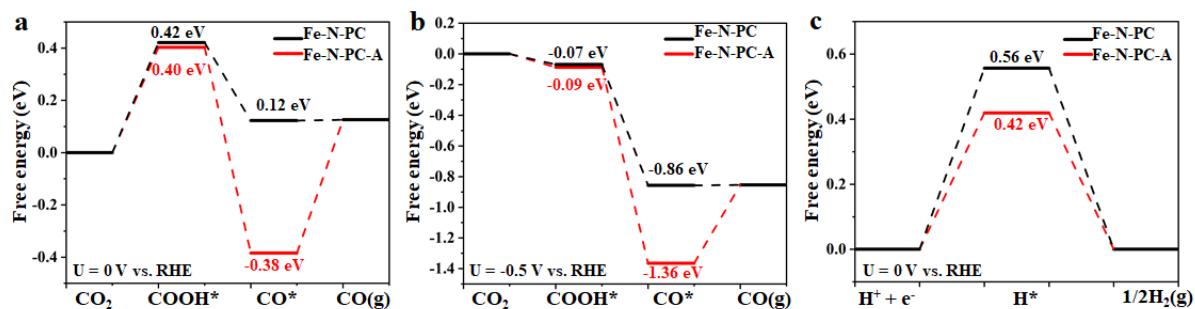


Figure 6. Free energy profiles for the CO₂RR at (a) 0 V (RHE) and (b) -0.5 V (RHE), and (c) for the HER at 0 V (RHE) on Fe-N-PC and Fe-N-PC-A catalysts.

In conclusion, we present a facile route to prepare Fe and N co-doped porous carbon nanospheres with the high density of atomic Fe-N₄-C active sites and a few Fe clusters using SiO₂ as template and protecting shell. The introduction of SiO₂ shell conduces to improve content of Fe (3.9 wt. %) and N (7.82 at. %) in final catalyst. The best performing Fe-N-PC catalyst exhibits outstanding catalytic activity for electrochemical reduction of CO₂ to CO result from high density of active sites. The poisoning experiment and selective chemical modification demonstrate that the active site for CO₂RR probably include both of the Fe-N₄ center and the surrounding carbon atoms. Comparative experiments and DFT simulations further confirm that the Fe clusters further promote the CO₂RR activity due to the low free energy of CO desorption from Fe-N₄-C active sites. The synthesis strategy and the exploration of active sites might contribute to the rational design of highly active M-N-C catalysts.

Notes

The authors declare no competing financial interest.

Supporting information

Supporting Information Available: Comparison between catalysts found in literature;

Selectivity results in catalyst of various catalysts;

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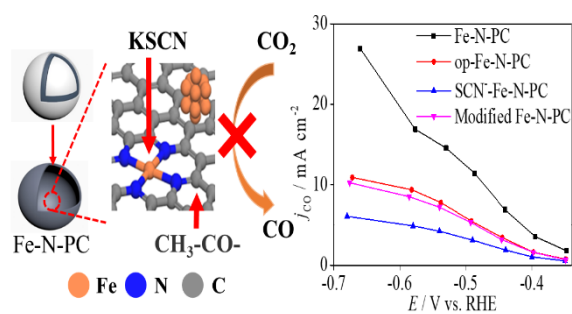
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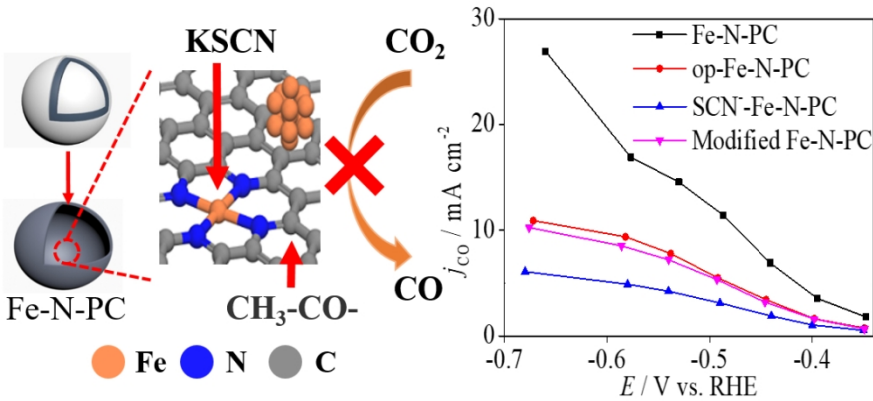
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TOC Graphic





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